

SITKA 1 NE, AK, USA

WMO: 703730

Lat: 57.0574N

Lon: 135.3266W

Elev: 24

StdP: 101.04

Time Zone: -9.00 (NAK)

Period: 05-19

WBAN: 25379

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-10.1	-7.6	-15.1	1.0	-6.9	-12.3	1.3	-5.1	2.1	6.0	1.8	5.5	0.2	N/A	0.596

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.7	19.8	14.9	18.1	14.3	16.8	13.6	15.7	18.4	15.1	17.1	14.5	16.1	0.8	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.6	10.4	15.9	14.1	10.1	15.5	13.6	9.8	15.2	44.0	18.7	42.0	17.2	40.5	16.1	19.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 1.6	1.3	1.2	DB	-12.3	24.0	3.2	2.0	-14.5	25.5	-16.4	26.7	-18.2	27.8	-20.5	29.3	(4)
(5)			WB	-12.1	17.2	3.0	1.2	-14.3	18.1	-16.1	18.8	-17.8	19.4	-19.9	20.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	6.4	1.4	0.9	1.9	4.9	8.2	10.8	13.0	13.3	10.9	6.7	2.7	1.2
	DBStd	5.32	3.78	3.45	3.26	2.14	2.22	1.96	1.69	1.47	1.90	2.39	3.44	3.54
	HDD10.0	1611	266	254	252	152	66	11	1	0	12	104	220	273
	HDD18.3	4373	523	488	511	402	316	225	167	157	224	361	470	531
	CDD10.0	280	1	0	0	1	8	36	92	102	38	2	0	0
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	5	0	0	0	0	1	1	3	1	0	0	0	0
	CDH26.7	1	0	0	0	0	0	0	1	0	0	0	0	0
Wind	WSAvg	0.3	0.3	0.3	0.4	0.4	0.4	0.3	0.3	0.3	0.3	0.3	0.4	0.4
Precipitation	PrecAvg	2218	207	160	153	125	117	93	104	169	286	352	236	214
	PrecMax	3123	400	395	327	284	255	206	242	496	527	783	426	384
	PrecMin	1527	41	6	22	29	31	23	42	44	100	161	42	57
	PrecStd	374	90	92	66	60	59	50	45	99	117	128	94	87
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.9	10.0	12.1	15.3	21.4	22.0	21.0	20.8	19.3	13.9	10.8	11.2
		MCWB	9.2	6.8	7.1	7.5	11.3	15.6	16.1	15.9	13.3	11.3	8.9	6.6
	2%	DB	8.6	8.6	9.5	12.2	16.4	18.2	19.2	19.2	16.8	12.2	9.6	8.4
		MCWB	7.2	6.0	6.1	7.1	10.4	14.3	15.2	15.2	13.1	10.5	8.7	6.3
	5%	DB	7.1	7.0	7.9	10.6	14.5	16.3	17.6	17.8	15.4	11.3	8.5	6.9
		MCWB	5.9	5.4	5.1	6.7	10.2	12.9	14.4	14.7	12.9	9.9	7.5	5.6
	10%	DB	5.9	5.7	6.4	9.3	12.8	14.8	16.1	16.5	14.3	10.4	7.4	5.6
		MCWB	5.1	4.6	4.6	6.1	9.5	11.9	13.7	14.2	12.8	9.2	6.7	4.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.0	7.9	8.8	9.1	12.4	16.2	16.5	16.5	15.5	12.4	9.9	8.0
		MCDB	12.2	8.9	12.3	11.1	16.6	21.2	20.3	19.0	16.5	12.9	10.0	8.8
	2%	WB	7.9	6.7	7.1	8.0	11.7	14.7	15.6	15.7	14.5	11.5	8.9	6.8
		MCDB	8.9	7.8	9.3	10.8	15.6	17.9	18.6	17.8	15.4	12.0	9.3	8.2
	5%	WB	6.6	5.8	6.1	7.3	10.8	13.6	14.8	15.2	13.8	10.5	7.9	5.7
		MCDB	7.4	6.8	7.9	10.1	13.7	15.9	16.9	17.0	14.8	11.1	8.4	6.8
	10%	WB	5.6	4.9	5.2	6.7	10.0	12.6	14.1	14.7	13.1	9.7	7.0	5.0
		MCDB	6.1	5.9	6.8	9.2	12.4	14.4	15.7	16.2	14.3	10.4	7.5	5.6
Mean Daily Temperature Range	5% DB	MDBR	3.9	5.2	6.2	7.5	7.4	6.5	5.4	5.7	6.0	5.3	4.1	3.9
		MCDBR	4.8	6.3	8.5	10.4	11.9	10.2	9.0	9.0	8.5	6.2	4.5	4.6
		MCWBR	3.4	4.0	5.4	5.8	6.8	5.6	5.1	5.4	5.5	4.7	3.7	3.4
	5% WB	MCDBR	4.5	5.0	7.7	8.2	9.0	8.4	7.2	6.7	6.7	4.7	4.2	4.4
		MCWBR	3.8	3.6	5.3	5.2	5.6	5.1	4.4	4.5	4.8	4.1	3.8	3.7
Clear-Sky Solar Irradiance	taub	0.299	0.312	0.331	0.395	0.372	0.369	0.381	0.360	0.354	0.339	0.304	0.280	
	taud	2.249	2.385	2.374	2.189	2.384	2.441	2.447	2.527	2.503	2.398	2.272	2.189	
	Ebn at Noon	570	748	828	813	865	872	852	850	800	698	556	476	
	Edh at Noon	51	68	88	124	109	104	102	88	79	67	49	41	
All-Sky Solar Radiation	RadAvg	0.38	1.06	2.17	3.64	4.62	4.74	4.01	3.46	2.22	1.18	0.51	0.27	
	RadStd	0.06	0.21	0.31	0.36	0.50	0.42	0.47	0.44	0.37	0.16	0.07	0.04	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (12 neighbors)	N/A	+1.11	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page